Lecture 2 Live Problems

Quantification. State variables and change.

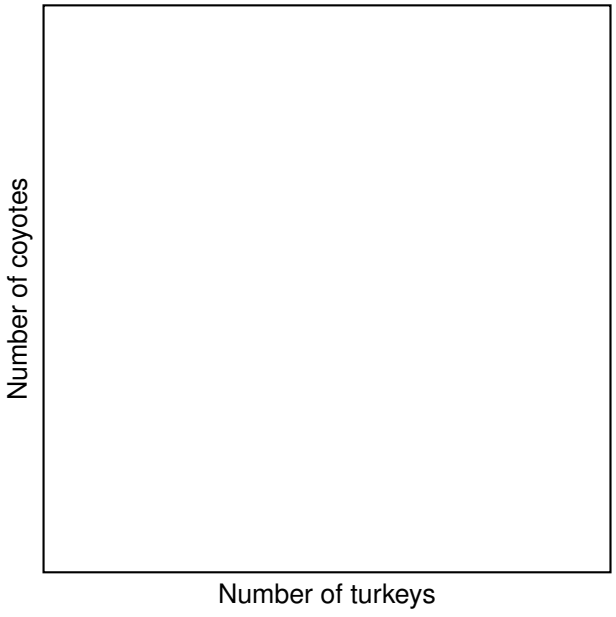
Points toward lab manual pages: **ST1, ST2, ST3, ST4** (section L1 S1)

1. Forrest Gump's hair. Describe your predictions for Forrest Gump's hair length H with a time series plot, given that

- he begins with a short 1 cm haircut;
- he ends 1200 days later with hair length 60 cm (about 2 feet, stretched).

2. What is the rate of Forrest Gump's hair growth, i.e., the rate of change of H? Use this to tabulate his hair length over the first 4 months.

t (months)	0	1	2	3	4
H (cm)	1				

3. Turkeys and coyotes, with numbers. Tabulate your predictions for the populations of turkeys and coyotes over 5 months, if there are currently 1000 turkeys and 20 coyotes. Plot these in a time series and as a trajectory in state space.

t (months)	0	1	2	3	4	5
T	1000					
C	20